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Effects of Exercise on Physiology

Introduction

In modern times, people are driven to maintain their physical performance to achieve better fitness. Healthy bodies are important to people for various reasons, ranging from aesthetic appearance to simply being in shape. Typically, a higher level of fitness or being in shape is associated with a better overall mood within an individual. Quality health derives from the combination of a good diet, exercise, sleep, and overall lifestyle. However, physical health places a rather large spotlight on exercise, whether that be aerobic or anaerobic.

Similar to exercise, the foundation of all health eventually comes down to breathing. Proper breathing can be the difference between a runner winning or losing a race. The ability to control one's breathing patterns is a subtle yet effective method to improve physical performance. If swimmers could learn to control their breathing underwater, their times would be significantly reduced. To continue, the comprehension of respiratory physiology is not just beneficial for athletes, but also for health professionals. Healthcare workers would be better able to monitor breathing levels and identify the origin of respiratory symptoms in a patient. Additionally, doctors, along with researchers, can accurately determine safe or even efficient rates of oxygen consumption. This has the potential to increase productivity in all of these different environments by simply understanding how to breathe efficiently. For these reasons, the results of this experiment should interest athletes, health professionals, and researchers.

Exercise pushes the body to initiate numerous processes in order to maintain homeostasis, which implies that exercise places physiological stress on the body. This homeostasis revolves around the lungs and heart. Changes that the body must compensate for include increased temperature, lack of oxygen, and increased carbon dioxide levels. During exercise, the human body is forced to regulate temperature, increase oxygen uptake, and expel carbon dioxide at a faster pace.

This experiment will serve to better understand the effects of exercise and recovery on various physiological factors within the body. In this study, it is hypothesized that exercise will increase heart rate, mean arterial pressure (slightly), carbon dioxide (CO_2) clearance, oxygen (O_2) consumption, and temperature. Whereas, hemoglobin (Hb) saturation will decrease. Conversely, recovery is hypothesized to decrease heart rate, MAP (slightly), CO_2 clearance, O_2 consumption, and temperature and return these values to their baseline. Whereas, Hb saturation will increase to its baseline value.

Methods

Initially, the target exercise heart rate is calculated out of safety for the runner because 80% of the maximal heart rate is considered a safe exercise heart rate and exceeding it can be dangerous. Heart rate can be used to assess exercise intensity and at 80% it would be at an intensity high enough to see a response. In this experiment, data was collected for heart rate and hemoglobin saturation (Hb Sat.) with the use of a pulse oximeter. Also, to calculate CO_2 clearance and O_2 consumption, exhaled tidal CO_2 and O_2 were determined with the gas analyzer setup and PowerLab. The runner is given mouthpieces in order to measure exhaled tidal oxygen (ETO_2) and exhaled tidal carbon dioxide (ETCO_2). Furthermore, body temperature was measured

with a laser thermometer and blood pressure was determined with the use of a sphygmomanometer. The blood pressure is then eventually used to calculate mean arterial pressure (MAP). The following equations were used to calculate MAP, O₂ consumption, and CO₂ clearance:

$$(\frac{2}{3}) \text{ Diastolic Blood Pressure} + (\frac{1}{3}) \text{ Systolic Blood Pressure} = \text{MAP}$$

$$(0.209 - \text{ETO}_2) \times \text{TV (L/breath)} \times \text{Resp. Rate (breath/min)} = \text{O}_2 \text{ Consumption (L O}_2 \text{ / min)}$$

$$\text{ETCO}_2 \times \text{TV (L/breath)} \times \text{Resp. Rate (breath/min)} = \text{CO}_2 \text{ Clearance (L CO}_2 \text{ / min)}$$

The exhaled tidal O₂ (ETO₂) is subtracted from 0.209, which represents atmospheric oxygen. This number is then multiplied to tidal volume (TV), which is normal breathing, and respiratory rate (breaths per minute), to get O₂ consumption (L O₂ / min). Moreover, the exhaled CO₂ (ETCO₂) is multiplied by the tidal volume and respiratory rate to obtain CO₂ clearance (L CO₂ / min).

Throughout the whole experiment for the rest, exercise, and recovery phases, all data was approximately collected at every 3 minute interval with their designated equipment. To begin, the control for the experiment, also known as the rest phase comes first. The rest phase is the control because the experiment's purpose is to observe exercise's effects in relation to a normal value. In the rest phase, the participant sits on a chair for 3 minutes twice. Following the rest stage, came the exercise portion in which the participant begins by running on a treadmill at 5 mph for 12 minutes. Incline was raised every 3 minutes starting at a 0 degree angle and ending at 10 degrees. After the runner steps off the treadmill, blood pressure was taken immediately. Lastly, in the recovery phase the participant walks for 3 minutes, then sits down for another 3 minutes. As the participant is sitting in the second 3 minute interval of the recovery phase, all measurements are taken once more. Some data was not collected during the exercise portion as it

was simply too difficult to measure variables such as blood pressure during motion with the sphygmomanometer and the stethoscope. Using the stethoscope during motion will make it hard to hear the sounds of Korotkoff (Waters et. al., 2022, p. 210).

Results

Figure 1: Changes in Heart Rate

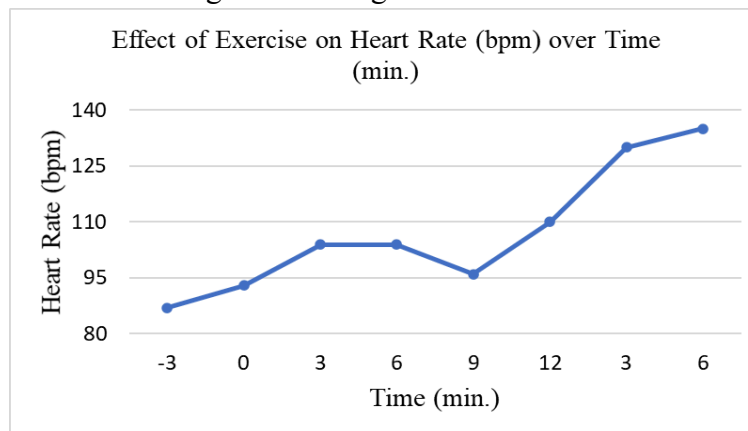


Figure 1 illustrates the change in heart rate over the rest, exercise, and recovery phases. -3 and 0 minutes represents three and six minutes into the rest period. 3, 6, 9, and 12 minutes represent the number of minutes the participant was into the exercise period. The last 3 and 6 minutes represent the number of minutes the participant was into the recovery period.

Figure 2: Changes in CO₂ Clearance

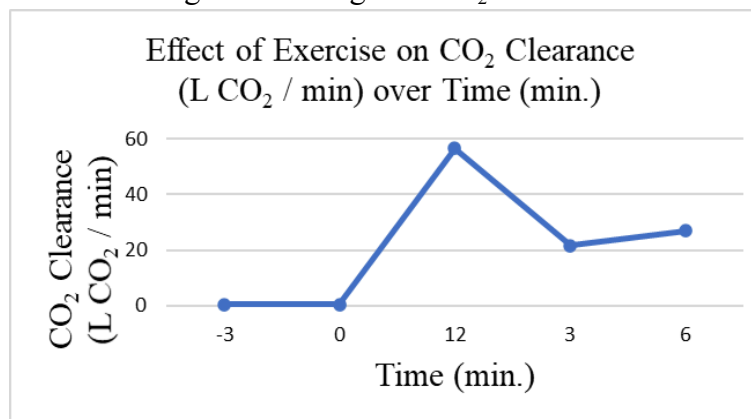


Figure 2 illustrates the change in CO₂ Clearance (L CO₂ / min) over the rest, exercise, and recovery phases. -3 and 0 minutes represents three and six minutes into the rest period. 12 minutes represents the number of minutes the participant was into the exercise period. The last 3 and 6 minutes represent the number of minutes the participant was into the recovery period.

Figure 3: Changes in O₂ Consumption

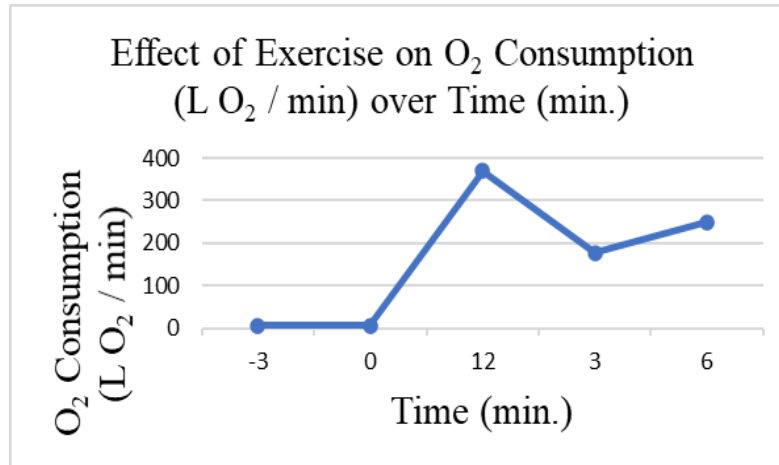


Figure 3 illustrates the change in O₂ Consumption (L O₂ / min) over the rest, exercise, and recovery phases. -3 and 0 minutes represents three and six minutes into the rest period. 12 minutes represents the number of minutes the participant was into the exercise period. The last 3 and 6 minutes represent the number of minutes the participant was into the recovery period.

Figure 4: Changes in Hemoglobin Saturation

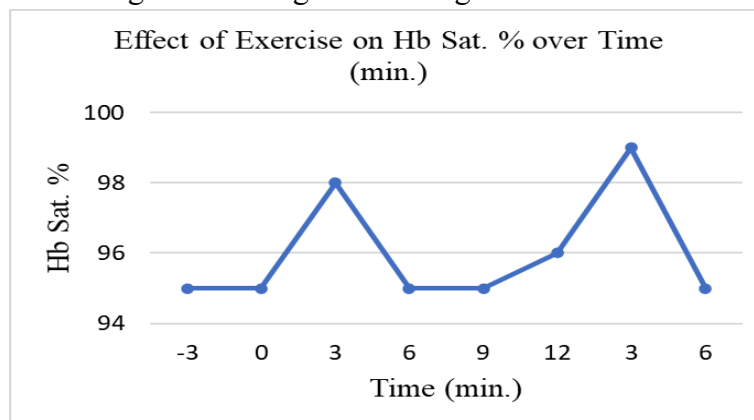


Figure 4 illustrates the change in hemoglobin saturation (%) over the rest, exercise, and recovery phases. -3 and 0 minutes represents three and six minutes into the rest period. 3, 6, 9, and 12 minutes represent the number of minutes the participant was into the exercise period. The last 3 and 6 minutes represent the number of minutes the participant was into the recovery period.

Figure 5: Changes in Body Temperature

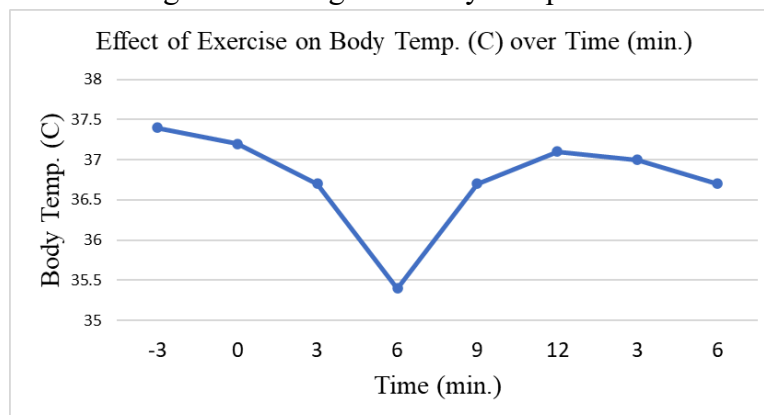


Figure 5 illustrates the change in body temperature (degrees Celsius) over the rest, exercise, and recovery phases. -3 and 0 minutes represents three and six minutes into the rest period. 3, 6, 9, and 12 minutes represent the number of minutes the participant was into the exercise period. The last 3 and 6 minutes represent the number of minutes the participant was into the recovery period.

Table 1: Changes in Mean Arterial Pressure (MAP)

Time (minutes)	-3	0	Soon as running stops	3	6
MAP (mmHg)	83	74	95	87	83

Table 1 shows the values for mean arterial pressure (mmHg) for a participant over the rest, exercise, and recovery phases. -3 and 0 minutes represents three and six minutes into the rest period. The last 3 and 6 minutes represent the number of minutes the participant was into the recovery period. MAP is calculated by adding ($\frac{2}{3}$) diastolic blood pressure and ($\frac{1}{3}$) systolic blood pressure together.

Overall, the key findings show that going from rest to exercise had a clear increase in heart rate, and this heart rate remained roughly similar excluding that it dropped to 96 bpm after 12 minutes of exercise. Moreover, the heart rate continued to rise during the recovery period. CO₂ clearance increased from the rest to exercise period, then dropped in the recovery phase. Similarly, O₂ consumption increased from rest to exercise and then dropped in the recovery period. To continue, hemoglobin saturation rose from resting to exercise, then dropped during the exercise period, and during recovery it rose and dropped once again. Body temperature decreased going from rest to exercise, then rose during the exercise period, and finally began to plateau during the recovery phase. Lastly, MAP rose during exercise and then returned to its baseline value in recovery. In all of these trends an increase or decrease is seen, however small it may be in value.

Discussion

In this experiment, the results for heart rate's recovery phase, body temperature, and Hb saturation were unexpected and did not support these expectations. Whereas, the results for CO₂ clearance, O₂ consumption, and MAP were expected and did support the hypothesis. According

to the findings, heart rate increased from rest to exercise as expected, but then continued to rise during recovery, which is illogical. This is because during exercise, runners require more oxygen in their muscles, meaning the heart has to pump faster to put more blood into systemic circulation. This is not consistent with the results since the heart should have pumped slower in recovery compared to during exercise. Moreover, although now there is more blood circulating within the body, the mean arterial pressure still only rises slightly, and this is consistent with the results. This is because vasodilation acts in skeletal muscle arterioles to lower arterial blood pressure, which cancels out some of the increase in blood pressure (Waters et. al., 2022, p. 200).

During exercise, CO_2 levels increase because CO_2 is a byproduct of cellular respiration. This raised the body's overall acidity in the blood. In an attempt to raise the pH and return the body's CO_2 levels back to normal, the respiratory rate increased. This homeostatic mechanism serves to expel CO_2 at a faster rate to maintain the body's natural pH. Moreover, this is consistent with the results for CO_2 clearance since there was an increase in $\text{L CO}_2 / \text{min}$ during exercise and a decrease in recovery. Similarly, since O_2 levels in the body were low during exercise, O_2 consumption increased in order to provide sufficient oxygen to the body and its tissues. To continue, heat is generated as a result of skeletal muscle contractions and cellular respiration, which in turn increases body temperature (Waters et. al., 2022, p. 200). This is not consistent with the data since body temperature should have increased from rest to exercise and not decreased. Lastly, Hb saturation decreases with exercise because oxygen is not connected as securely to hemoglobin. Also, increased temperature and CO_2 , and decreased pH, are all factors associated with exercise that can increase oxygen unloading from hemoglobin in exercising muscles (Waters et. al., 2022, p. 204). This process shifts the hemoglobin dissociation curve down and is helpful because more oxygen is now available for the body's tissues. The results for

Hb saturation are not consistent with this physiological mechanism because Hb saturation rose from rest to exercise and never dropped below its resting value.

Overall, the experimental results for CO₂ clearance, O₂ consumption, and MAP are consistent with the hypothesis, whereas the data for heart rate, body temperature, and Hb saturation are not. In addition, it was observed that exercise has a significant impact on most of these physiological factors. Possible sources of error in this experiment include impurities on the participant's fingers such as dirt, nail polish, or other debris that could have interfered with the pulse oximeter. Also, values for the exercise stage may be off due to not properly measuring the variables since it was difficult to do so mid-exercise. Additionally, a lack of experience from the data collectors could have led to determining misleading values for the various measurement tools. Those who wish to replicate this study in the future should perform more trials and have more runners to increase reliability and accuracy. Also, an information session for the runners and data collectors on how to use the measurement tools prior to the experiment would be beneficial. All of these suggestions will help to improve the experiment in the future and reduce the risk of error.

This study is important for doctors, researchers, and athletes. More so doctors than anyone else, because detecting abnormal behavior in any of the physiological factors in the experiment could save a life. Athletes also have much to gain from understanding oxygen consumption and carbon dioxide disposal. Proper training can help athletes maximize their oxygen intake and improve overall performance. Comprehension of exercise physiology is significant to healthcare and overall physical performance because it allows individuals to have conscious awareness of their physiological mechanisms.

References

Waters, J. R., Tomicek, N. J., Neisser, J. P., Squyres, N. S., Waters, D. N., & Hackenbrack, N. M. (2022). *Laboratory Manual: BIOL 164: Human Anatomy & Physiology II* (Second Edition). Plymouth, MI: Hayden-McNeil Lab Solutions.